



Recitation 15

Yuchao Chen, Mingtao Huang

{cyc20, huangmt19}@mails.tsinghua.edu.cn

Tsinghua University

June 9/10/11, 2021

- Date: June 16, 2021
- Time: 8:00-10:00 (GMT+08:00, Beijing time)
- Location: Lecture hall, Department Building of School of Architecture (建馆报告厅)
- Coverage: Lecture 1 to 14
- Additional information:
 - Closed-book exam, with one piece of A4 “cheat sheet” available (Two-sided)
 - Calculator **NOT** necessary
- Account for 40% in final grade
- Please get well-prepared! Good luck!
- Q&A session: 13:00-17:00, Jun. 14 (Mon.), 10-208, Rohm Building

1. Homework

2. Recitation

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1. Each of n packages is loaded independently onto either a red truck (with probability p) or onto a green truck (with probability $1 - p$). Let R be the total number of items selected for the red truck and let G be the total number of items selected for the green truck.
 - (c) Evaluate the probability that at least one truck ends up with a total of exactly one package.



1. Each of n packages is loaded independently onto either a red truck (with probability p) or onto a green truck (with probability $1 - p$). Let R be the total number of items selected for the red truck and let G be the total number of items selected for the green truck.
- (c) Evaluate the probability that at least one truck ends up with a total of exactly one package.

Note: $n=1$ and $n=2$ case should be separated.

Let B be the event that at least one truck ends up with a total of exactly one package.

$$\mathbf{P}(B) = \begin{cases} 1 & \text{if } n = 1 \\ 2p(1 - p) & \text{if } n = 2 \\ n(1 - p)^{n-1}p + np^{n-1}(1 - p) & \text{if } n = 3, 4, \dots \end{cases}$$

3. Suppose a bank has 2 tellers. When you enter the bank, you find that both tellers are busy serving other customers, and there are no other customers in queue. The service times of the two tellers are independent distributed exponential random variables, with parameters λ_1 and λ_2 respectively. What is the probability that you will be the last to leave?

3. Suppose a bank has 2 tellers. When you enter the bank, you find that both tellers are busy serving other customers, and there are no other customers in queue. The service times of the two tellers are independent distributed exponential random variables, with parameters λ_1 and λ_2 respectively. What is the probability that you will be the last to leave?

We can view the two tellers as two Poisson processes with parameters λ_1 and λ_2 respectively. Here a success means a teller finishes serving a customer.

We can then merge these two Poisson processes together, resulting in a merged Poisson process with parameter $\lambda_1 + \lambda_2$. When the merged process successes, the success comes from the first process with probability $\lambda_1/(\lambda_1 + \lambda_2)$, which means that the customer at the first teller leaves first with probability $\lambda_1/(\lambda_1 + \lambda_2)$.

And because of memoryless property, when the first customer leaves, the situation comes back to where we started.

Therefore,

$$\begin{aligned}\mathbf{P}(\text{you leave last}) &= \mathbf{P}(\text{A leaves first})\mathbf{P}(\text{you leave last} \mid \text{A leaves first}) \\ &\quad + \mathbf{P}(\text{B leaves first})\mathbf{P}(\text{you leave last} \mid \text{B leaves first}) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_2}{\lambda_1 + \lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_1}{\lambda_1 + \lambda_2} \\ &= \frac{2\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2}\end{aligned}$$

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2. Recitation

1. (Lecture 15) We toss n times a biased coin whose probability of heads, denoted by q , is the value of a random variable Q with given mean μ and positive variance σ^2 . Let X_i be a Bernoulli variable that models the outcomes of the i th toss. We assume that $X_1, \dots, X_n$ are conditionally independent, given $Q = q$. Let X be the number of heads obtained in the n tosses.

- (a) Use the law of iterated expectation to find $\mathbf{E}[X_i]$ and $\mathbf{E}[X]$.
- (b) Find $\text{cov}(X_i, X_j)$. Are $X_1, \dots, X_n$ independent?

- (a) We have, from the law of iterated expectation and the fact $\mathbf{E}[X_i|Q] = Q$,

$$\mathbf{E}[X_i] = \mathbf{E}[\mathbf{E}[X|Q_i]] = \mathbf{E}[Q] = \mu.$$

Since $X = X_1 + \cdots + X_n$, it follows that

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \cdots + \mathbf{E}[X_n] = n\mu.$$

(b) We have, for $i \neq j$,

$$\mathbf{E}[X_i X_j | Q] = \mathbf{E}[X_i | Q] \mathbf{E}[X_j | Q] = Q^2,$$

and

$$\mathbf{E}[X_i X_j] = \mathbf{E}[\mathbf{E}[X_i X_j | Q]] = \mathbf{E}[Q^2].$$

Thus

$$\text{cov}(X_i X_j) = \mathbf{E}[X_i X_j] - \mathbf{E}[X_i] \mathbf{E}[X_j] = \mathbf{E}[Q^2] - \mu^2 = \sigma^2.$$

Since $\text{cov}(X_i, X_j) > 0$, $X_1, \dots, X_n$ are not independent.

2. (Final 2019) Let $X_1, X_2, \dots, X_n$ be independent variables. For each $i \in \{1, 2, \dots, n\}$, X_i follows a **standard normal distribution**. Define $Y = X_1^2 + X_2^2 + \dots + X_n^2$.

- (a) Find $\mathbf{E}[Y]$ and $\text{var}(Y)$.
- (b) Suppose that $n = 100$. Find a good approximation for the probability that $\mathbf{P}(Y \geq 400)$. Give your answer in terms of the standard normal CDF $\Phi(x)$.

(a) If X is $\mathcal{N}(0, 1)$, then the moment generating function of X^2 is

$$M_{X^2}(s) = \mathbf{E}[e^{sx^2}] = \int_{-\infty}^{\infty} e^{sx^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = \frac{1}{\sqrt{1-2s}}.$$

If $s < 1/2$, it follows that

$$M_Y(s) = \prod_{j=1}^n M_{X_j^2}(s) = \frac{1}{(1-2s)^{n/2}}.$$

Hence, we have

$$\begin{aligned}\mathbf{E}[Y] &= \left. \frac{\partial}{\partial s} M_Y(s) \right|_{s=0} = n \\ \mathbf{E}[Y^2] &= \left. \frac{\partial^2}{\partial s^2} M_Y(s) \right|_{s=0} = n(n+2).\end{aligned}$$

Consequently, $\text{var}(Y) = 2n$.

- (b) Suppose that $n = 100$, we approximate Y as a Gaussian random variable $\mathcal{N}(100, 200)$ using Central limit theorem. Thus,

$$\mathbf{P}(Y \geq 400) = 1 - \Phi\left(\frac{400 - 100}{\sqrt{200}}\right) = \Phi(-15\sqrt{2}).$$

3. (Final 2019) A fair coin is tossed n times, showing heads H_n times and tails T_n times. Let $S_n = H_n - T_n$.

(a) Find the transform of S_n .

(b) Prove that

$$\mathbf{P}(S_n > an)^{1/n} \leq 1/\sqrt{(1+a)^{1+a}(1-a)^{1-a}}, \quad \text{if } 0 < a < 1.$$

Hint: Chernoff inequality.

(c) What if $a > 1$?

- (a) Define the outcome of the i th toss as X_i , which takes value of 1 when it is a head while takes value of -1 otherwise. Thus

$$S_n = X_1 + \cdots + X_n.$$

Therefore,

$$M_{S_n}(s) = (M(X_i))^n = \left(\frac{e^s + e^{-s}}{2} \right)^n.$$

(b) Using the Chernoff inequality, we can obtain

$$\mathbf{P}(S_n \geq an) \leq e^{-\phi_{S_n}(an)},$$

where

$$\phi_{S_n}(an) = \max_{s \geq 0} \{san - \ln M_{S_n}(s)\},$$

Thus we have

$$\mathbf{P}(S_n \geq na)^{1/n} \leq e^{-\max_{s \geq 0} \{sa - \ln((e^s + e^{-s})/2)\}}$$

The right side has a minimum when $e^s = \sqrt{(1+a)/(1-a)}$.

(b) Thus, we get

$$\begin{aligned}sa - \ln \frac{e^s + e^{-s}}{2} &= a \ln \sqrt{\frac{1+a}{1-a}} - \ln \frac{\sqrt{\frac{1+a}{1-a}} + \sqrt{\frac{1-a}{1+a}}}{2} \\&= \ln \sqrt{\frac{(1+a)^a}{(1-a)^a}} - \ln \frac{1}{\sqrt{(1-a)(1+a)}} \\&= \ln \sqrt{\frac{(1+a)^{a+1}}{(1-a)^{a-1}}}\end{aligned}$$

Therefore,

$$\mathbf{P}(S_n \geq na)^{1/n} \leq e^{-\max_{s \geq 0} \{sa - \ln((e^s + e^{-s})/2)\}} = \sqrt{\frac{(1-a)^{a-1}}{(1+a)^{a+1}}}$$

(c) If $a > 1$, then $\mathbf{P}(S_n > an)^{1/n} = 0$. (We only toss n times, so S_n cannot be larger than n .)

4. (Final 2020) Let X and Y be two independent variables such that $X \sim \mathcal{N}(\mu, 1)$ and $Y \sim \mathcal{N}(0, 1)$.

(a) Find the PDF of $Z = \min(X, Y)$.

(b) For each $t \in \mathbb{R}$, calculate $\mathbf{P}(\max(X, Y) - \min(X, Y) \geq t)$.

Give all your answers in terms of the standard normal CDF $\Phi(x)$ and its first order derivative $\Phi'(x)$.

(a)

$$\begin{aligned}F_Z(z) &= 1 - \mathbf{P}(Z \geq z) \\&= 1 - \mathbf{P}(X \geq z)\mathbf{P}(Y \geq z) \\&= 1 - (1 - \mathbf{P}(X \leq z))(1 - \mathbf{P}(Y \leq z)) \\&= \mathbf{P}(X \leq z) + \mathbf{P}(Y \leq z) - \mathbf{P}(X \leq z)\mathbf{P}(Y \leq z) \\&= \Phi(z - \mu) + \Phi(z) - \Phi(z - \mu)\Phi(z)\end{aligned}$$

$$\begin{aligned}f_Z(z) &= \frac{d}{dz}F_Z(z) \\&= \Phi'(z - \mu) + \Phi'(z) - \Phi'(z - \mu)\Phi(z) - \Phi(z - \mu)\Phi'(z)\end{aligned}$$

(b) For $t \leq 0$, $\mathbf{P}(\max(X, Y) - \min(X, Y) \geq t) = 1$.

For $t > 0$, we define a new random variable $U = X - Y$. U satisfies

$$|U| = \max(X, Y) - \min(X, Y)$$

Since X, Y are two independent Gaussian random variables, the linear combination U is also a Gaussian random variable and $U \sim \mathcal{N}(\mu, 2)$. Therefore,

$$\begin{aligned} & \mathbf{P}(\max(X, Y) - \min(X, Y) \geq t) \\ &= \mathbf{P}(|U| \geq t) \\ &= 1 - \mathbf{P}(-t \leq U \leq t) \\ &= 1 - \mathbf{P}\left(\frac{-t - \mu}{\sqrt{2}} \leq \frac{U - \mu}{\sqrt{2}} \leq \frac{t - \mu}{\sqrt{2}}\right) \\ &= 1 - \left(\Phi\left(\frac{t - \mu}{\sqrt{2}}\right) - \Phi\left(\frac{-t - \mu}{\sqrt{2}}\right)\right) \end{aligned}$$

5. (Final 2019 & 2020) A store opens at $t = 0$ and potential customers arrive in a Poisson manner at an average arrival rate of λ potential customers per hour. As long as the store is open, and independently of all other events, each particular potential customer becomes an actual customer with probability p . The store closes as soon as ten actual customers have arrived.

- (a) What is the probability that the fifth potential customer to arrive becomes the third actual customer?
- (b) Find the process for the arrival of actual customers and its parameter.
- (c) What is the PDF and expected value for L , the duration of the interval from store opening to store closing?
- (d) Considering ten actual customers have arrived, what is the probability that no two actual customers arrive within τ time units of each other?

- (a) Since a given potential customer becomes an actual customer with probability p , we are dealing with a binomial probability, and the probability of exactly 3 of the first 5 potential customers being actual customers is given by

$$\mathbf{P}(A) = \binom{4}{2} p^3 (1 - p)^2.$$

- (b) The process of incoming customers is being randomly split into two independent Poisson processes. The process for the arrival of actual customers is with rate $p\lambda$.

- (c) The distribution of $L = T_1 + T_2 + \dots + T_{10}$ is the 10th order Erlang PDF,

$$f_L(l) = \frac{(p\lambda)^{10} l^9 e^{-(p\lambda)l}}{9!}, \quad l \geq 0.$$

It follows that the expected value of L is $10/p\lambda$.

- (d) The probability of no two actual customers arriving within τ time units of each other is equivalent to the probability of all nine independent interarrival times, separating the ten actual customers, being at least τ time units apart. Hence,

$$\mathbf{P}(B) = \mathbf{P}(T_1 \geq \tau)^9 = e^{-9p\lambda\tau}.$$